

Experiment 1 — Installation of Hypervisors and Initiation of VMs with Image File

Cloud Computing Lab

Name	Deep Shekhar Halder
Roll Number	UG/02/BTCSE/2023/063
Course	B.Tech CSE, Adamas University
Subject	Cloud Computing Theory and Lab
Faculty	Anusuya Bera, Assistant Professor (CSE, SOET)
Experiment	Lab 1 — Hypervisor Installation and VM Setup

1. Objective

Install and configure an open-source hypervisor and create/instantiate a Virtual Machine (VM) using an existing image file. Demonstrate the successful installation and execution of the VM.

2. Tools Used

- Hypervisor: Oracle VirtualBox
- Guest OS: Ubuntu 26.04 (Desktop, 64-bit)
- Host OS: Windows

3. VM Configuration

- VM Name: Ubuntu-CloudLab
- Base Memory (RAM): 4096 MB
- Processors: 2 CPUs
- Hard Disk Size: 25 GB
- Graphics Controller: VBoxSVGA
- Network Adapter: NAT

4. Steps Performed

1. Downloaded and installed Oracle VirtualBox on the host machine.
2. Downloaded the Ubuntu Desktop (64-bit) ISO image.
3. Created a new VM in VirtualBox and configured CPU, RAM, storage, and network.
4. Attached the Ubuntu ISO to the VM's optical drive.
5. Started the VM and booted into the Ubuntu installer.
6. Completed the guest OS installation.
7. Verified the VM was running successfully by logging into the desktop.
8. Opened a terminal inside the VM and ran the following commands:

```
sudo apt update
sudo apt install -y build-essential linux-headers-$(uname -r)
```

5. Screenshots

5.1 VM Hardware Configuration

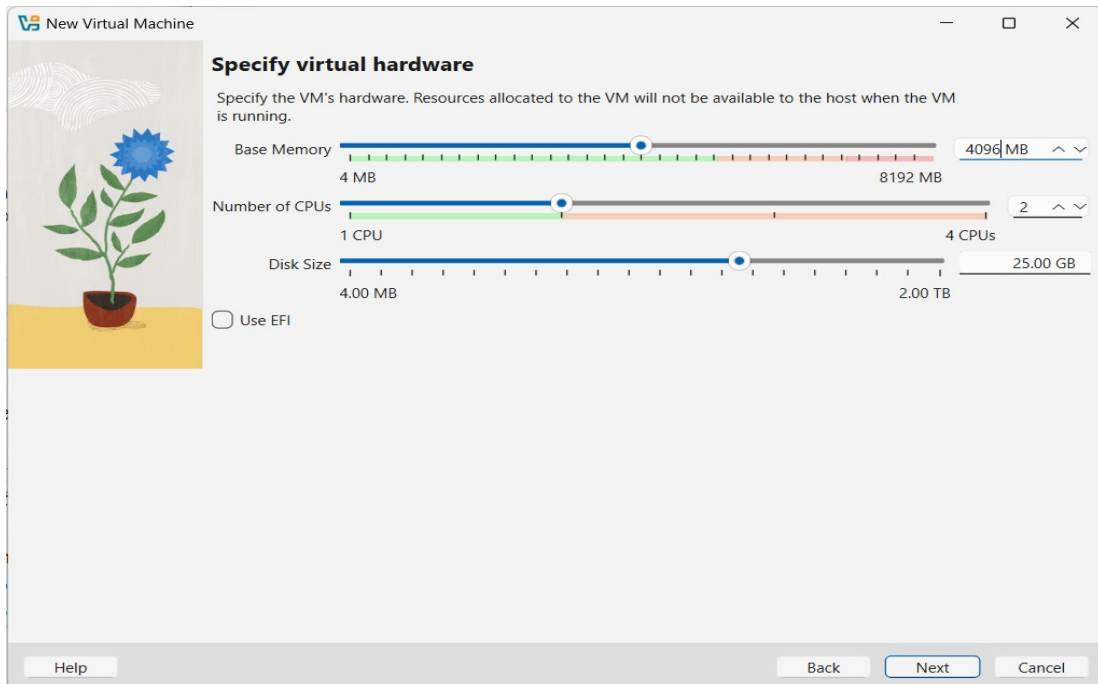


Figure 1: Specifying Base Memory, CPUs, and Disk Size during VM creation.

5.2 VM Configuration Summary

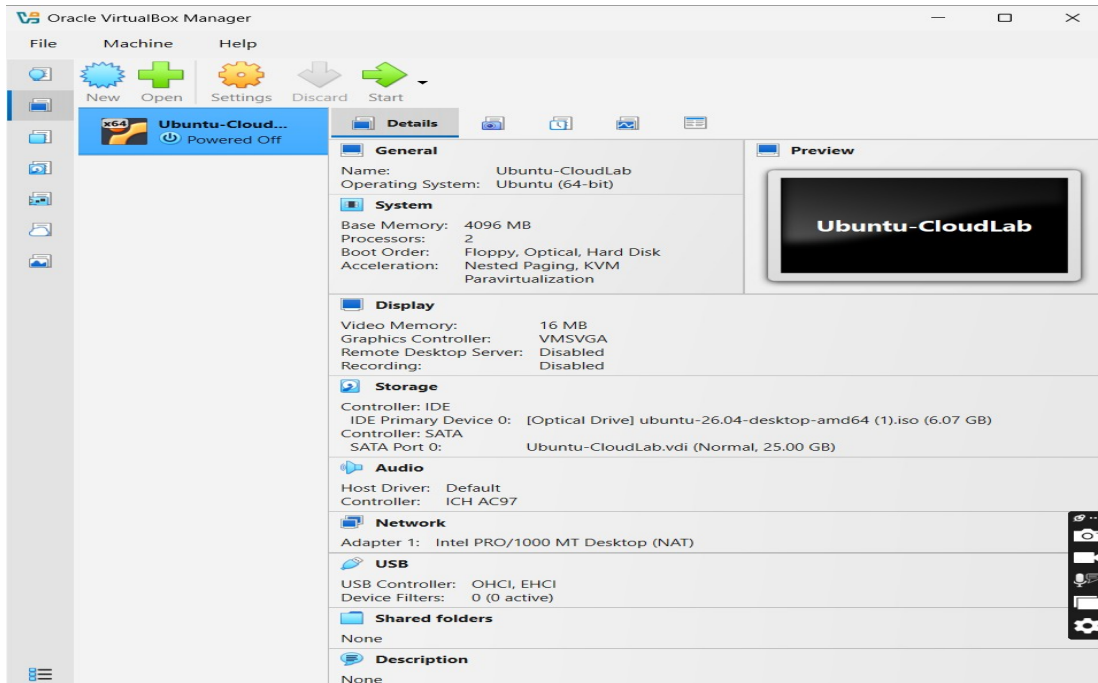


Figure 2: Final VM details — 4096 MB RAM, 2 CPUs, 25 GB disk, Ubuntu ISO attached.

5.3 Ubuntu Installer

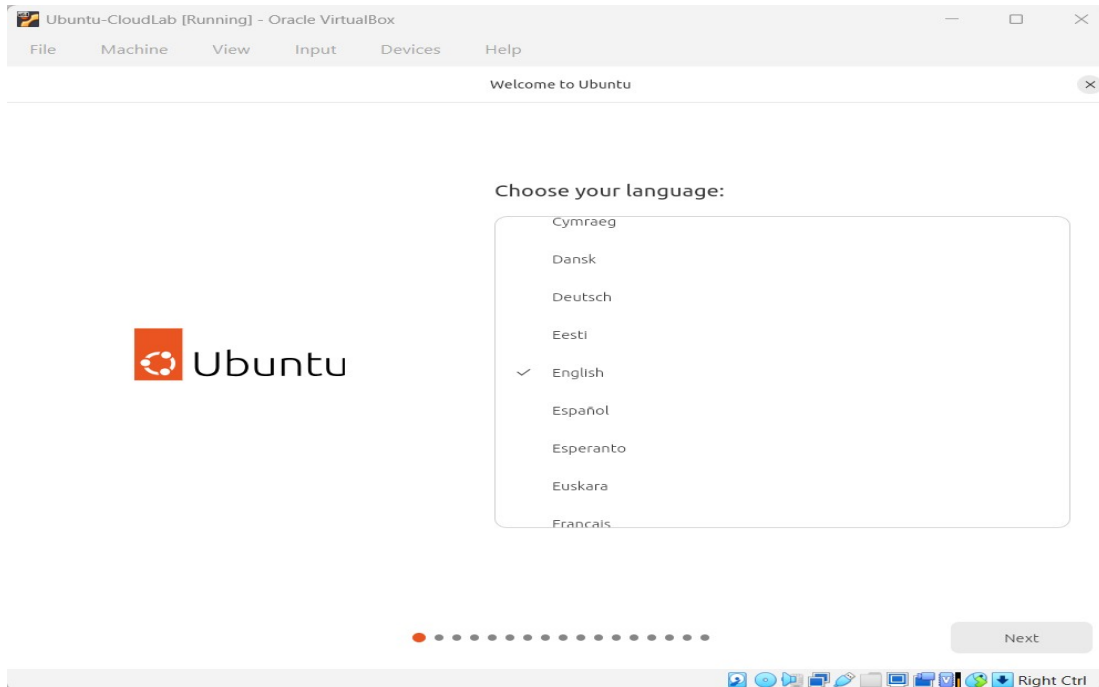


Figure 3: Ubuntu installer welcome screen after successful boot.

5.4 Running Ubuntu Desktop

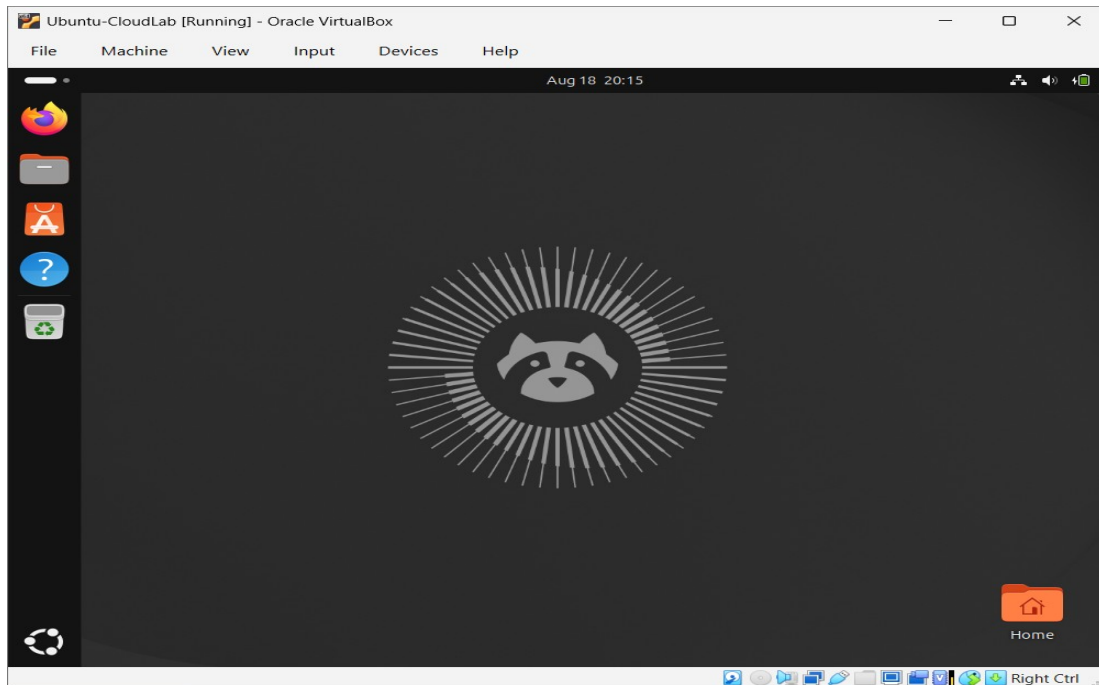


Figure 4: Ubuntu desktop running successfully inside the VM, confirming the installation.

5.5 Terminal Access and sudo Authentication

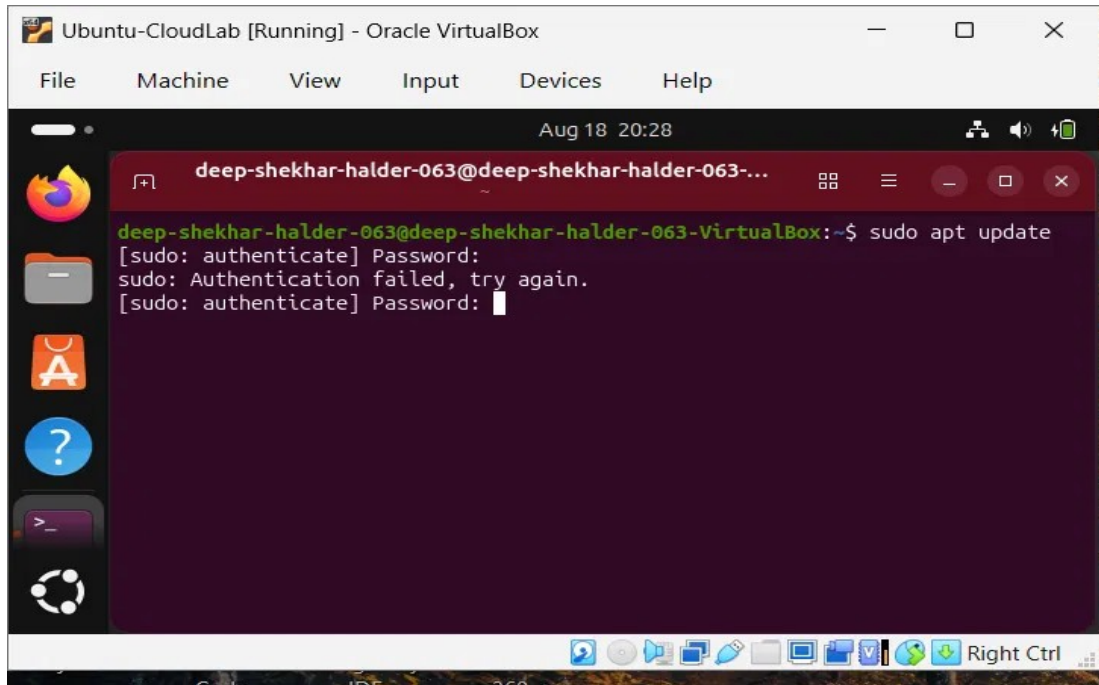


Figure 5: Terminal opened inside the Ubuntu VM, running 'sudo apt update'.

5.6 Package List Updated

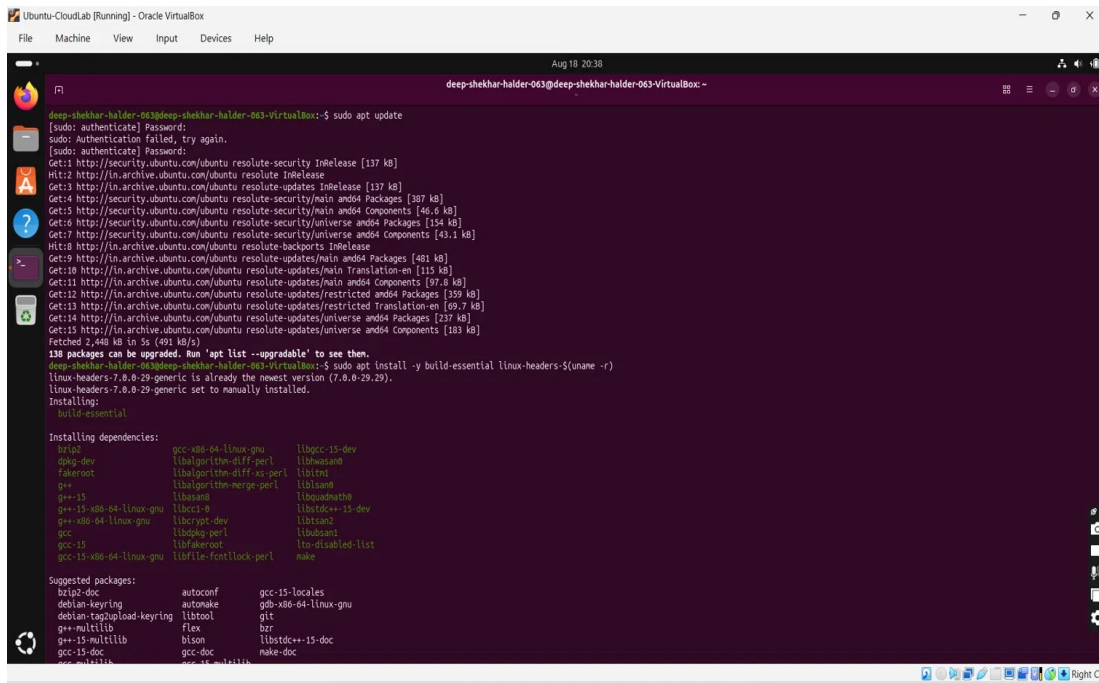
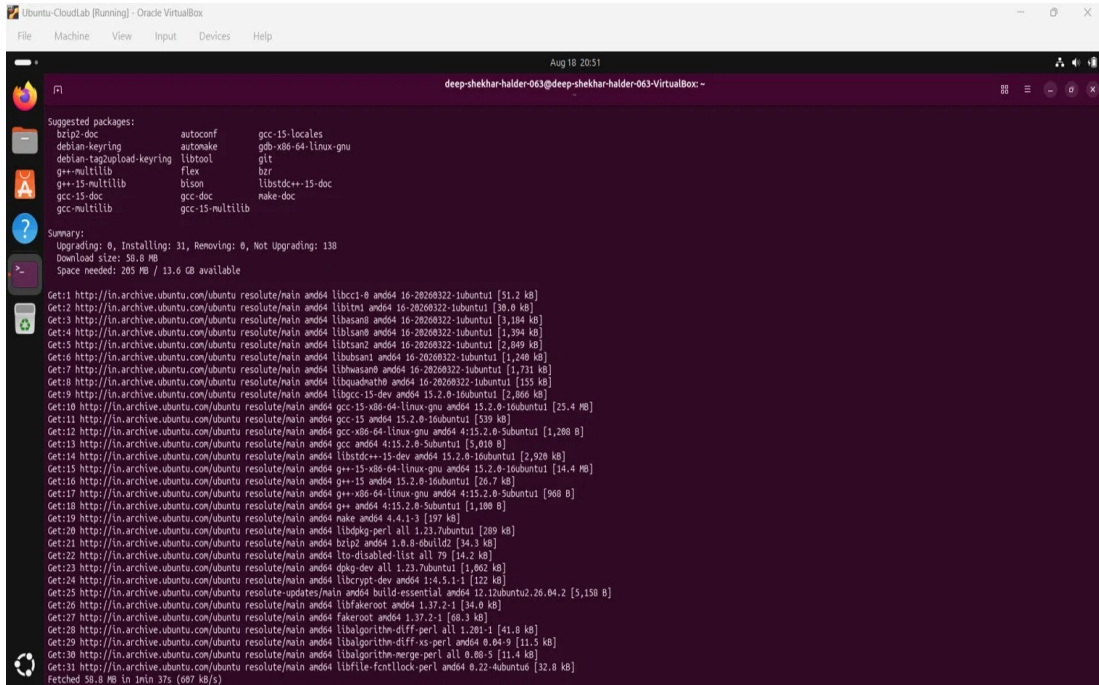


Figure 6: 'sudo apt update' completed; 'sudo apt install' started for build-essential and kernel headers.

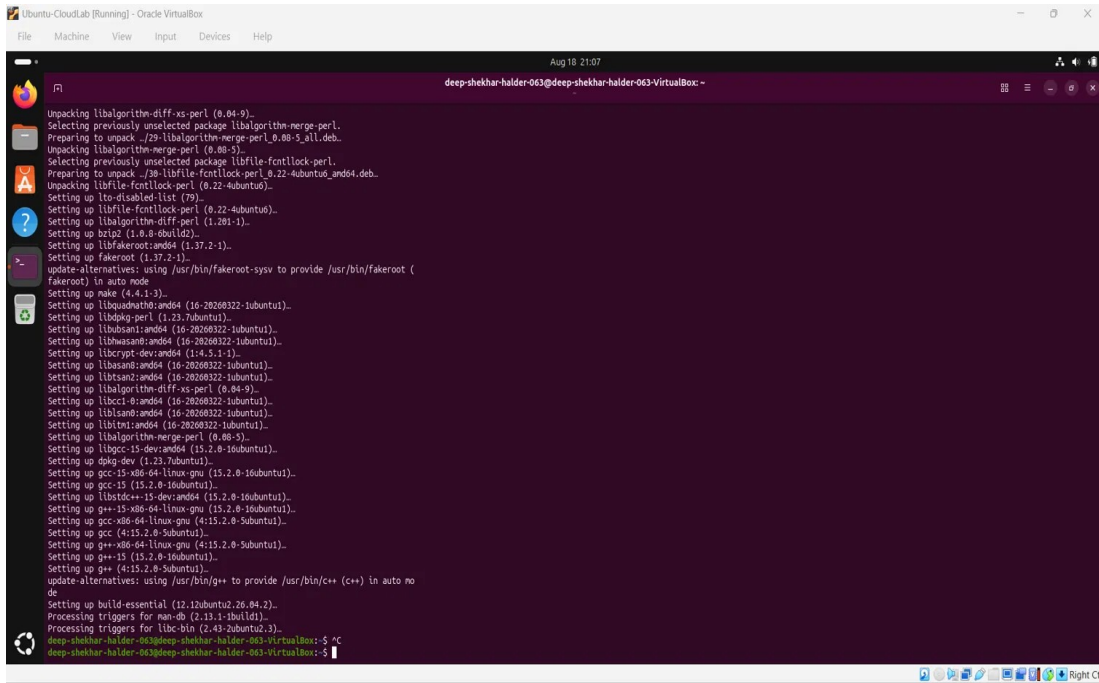
5.7 Installing Dependencies



```
deep-shekar-halder-063@deep-shekar-halder-063-VirtualBox: ~  
Suggested packages:  
  bzr2-doc      autoconf      gcc-15-locales  
  debian-keyring  autocow      g++-x86-64-linux-gnu  
  debian-tagload-keyring  libtool      g++  
  g++-multilib   flex          bzip2  
  gcc-15-doc     bison        libstdc++-15-doc  
  gcc-multilib   gcc-doc       make-doc  
  gcc-15-multilib  
  
Summary:  
Upgrading: 0, Installing: 31, Removing: 0, Not Upgrading: 138  
Download sizes: 58.8 MB  
Space needed: 205 MB / 13.6 GB available  
  
Get:1 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libcc1-0 amd64 16-20260322-1ubuntu1 [51.2 kB]  
Get:2 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libitm1 amd64 16-20260322-1ubuntu1 [38.0 kB]  
Get:3 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libsan0 amd64 16-20260322-1ubuntu1 [3,104 kB]  
Get:4 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libsan0 amd64 16-20260322-1ubuntu1 [3,394 kB]  
Get:5 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libsan2 amd64 16-20260322-1ubuntu1 [2,849 kB]  
Get:6 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libubsan1 amd64 16-20260322-1ubuntu1 [1,240 kB]  
Get:7 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libhwasm0 amd64 16-20260322-1ubuntu1 [1,231 kB]  
Get:8 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libquadmath0 amd64 16-20260322-1ubuntu1 [155 kB]  
Get:9 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libgcc-15-dev amd64 15.2.0-16ubuntu1 [2,866 kB]  
Get:10 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 gcc-15-x86-64-linux-gnu amd64 15.2.0-16ubuntu1 [25.4 MB]  
Get:11 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 gcc-15 amd64 15.2.0-16ubuntu1 [539 kB]  
Get:12 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 gcc-x86-64-linux-gnu amd64 4:15.2.0-Subunit [1,260 B]  
Get:13 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 gcc amd64 4:15.2.0-Subunit [5,018 B]  
Get:14 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libstdc++-15-dev amd64 15.2.0-16ubuntu1 [2,928 kB]  
Get:15 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 g++-15-x86-64-linux-gnu amd64 15.2.0-16ubuntu1 [14.4 MB]  
Get:16 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 g++-15 amd64 15.2.0-16ubuntu1 [26.9 kB]  
Get:17 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 g++-x86-64-linux-gnu amd64 4:15.2.0-Subunit [968 B]  
Get:18 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 g++ amd64 4:15.2.0-Subunit [1,100 B]  
Get:19 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 make amd64 4.4.1-3 [197 kB]  
Get:20 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libdpkg-dev all 1.23.7ubuntu1 [289 kB]  
Get:21 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 bzip2 amd64 1.0.8-0build2 [34.3 kB]  
Get:22 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 lto-disabled-list all 79 [14.2 kB]  
Get:23 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 dpkg-dev all 1.23.7ubuntu1 [1,662 kB]  
Get:24 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libcrypt-dev amd64 1:4.5.1-1 [122 kB]  
Get:25 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 build-essential amd64 12.12ubuntu2.26.04.2 [5,198 B]  
Get:26 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libfakeroot amd64 1.37.2-1 [34.0 kB]  
Get:27 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 fakeroot amd64 1.37.2-1 [68.3 kB]  
Get:28 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libalgorithm-diff-perl all 1.201-1 [41.8 kB]  
Get:29 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libalgorithm-diff-x86-perl amd64 0.04-9 [11.5 kB]  
Get:30 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libalgorithm-merge-perl all 0.08-5 [11.4 kB]  
Get:31 http://in.archive.ubuntu.com/ubuntu resolve/main amd64 libfile-fcntllock-perl amd64 0.22-4ubuntu2 [32.8 kB]  
Fetched 58.8 MB in 37s (687 kB/s)
```

Figure 7: Downloading build-essential and its dependencies.

5.8 Installation Completed Successfully



```
deep-shekar-halder-063@deep-shekar-halder-063-VirtualBox: ~  
Unpacking libalgorithm-diff-x86-perl (0.04-9).  
Selecting previously unselected package libalgorithm-merge-perl.  
Preparing to unpack ./29-libalgorithm-merge-perl_0.08-5_all.deb.  
Unpacking libalgorithm-merge-perl (0.08-5).  
Selecting previously unselected package libfile-fcntllock-perl.  
Preparing to unpack ./38-libfile-fcntllock-perl_0.22-4ubuntu2_amd64.deb.  
Unpacking libfile-fcntllock-perl (0.22-4ubuntu2).  
Setting up lto-disabled-list (79).  
Setting up libfile-fcntllock-perl (0.22-4ubuntu2).  
Setting up libalgorithm-diff-perl (1.201-1).  
Setting up bzip2 (1.0.8-0build2).  
Setting up libfakeroot (1.37.2-1).  
Setting up fakeroot (1.37.2-1).  
update-alternatives: using /usr/bin/fakeroot-sysv to provide /usr/bin/fakeroot ( fakeroot) in auto mode  
Setting up make (4.4.1-3).  
Setting up libquadmath0:amd64 (16-20260322-1ubuntu1).  
Setting up libdpkg-perl (1.23.7ubuntu1).  
Setting up libubsan1:amd64 (16-20260322-1ubuntu1).  
Setting up libhwasm0:amd64 (16-20260322-1ubuntu1).  
Setting up libsan2:amd64 (16-20260322-1ubuntu1).  
Setting up libcc1-0:amd64 (16-20260322-1ubuntu1).  
Setting up libsan0:amd64 (16-20260322-1ubuntu1).  
Setting up libitm1:amd64 (16-20260322-1ubuntu1).  
Setting up libgcc-15-dev:amd64 (15.2.0-16ubuntu1).  
Setting up libalgorithm-merge-perl (0.08-5).  
Setting up dpkg-dev (1.23.7ubuntu1).  
Setting up gcc-15-x86-64-linux-gnu (15.2.0-16ubuntu1).  
Setting up gcc-15 (15.2.0-16ubuntu1).  
Setting up libstdc++-15-dev:amd64 (15.2.0-16ubuntu1).  
Setting up g++-15-x86-64-linux-gnu (15.2.0-16ubuntu1).  
Setting up gcc-x86-64-linux-gnu (4:15.2.0-Subunit).  
Setting up gcc (4:15.2.0-Subunit).  
Setting up g++-15 (15.2.0-16ubuntu1).  
Setting up g++ (4:15.2.0-Subunit).  
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode  
Setting up build-essential (12.12ubuntu2.26.04.2).  
Processing triggers for man-db (2.13.1-1build1).  
Processing triggers for libc-bin (2.43-2ubuntu2.3).  
deep-shekar-halder-063@deep-shekar-halder-063-VirtualBox: ~  
deep-shekar-halder-063@deep-shekar-halder-063-VirtualBox: ~
```

Figure 8: build-essential and all dependencies set up successfully, returning to the shell prompt.

6. Conclusion

Oracle VirtualBox was successfully installed and used to create an Ubuntu virtual machine with the specified CPU, RAM, storage, and network configuration. The VM was booted, the guest OS installation was completed, and the VM was verified to be running successfully. The required build tools and Linux kernel headers were installed inside the VM without errors, confirming that the environment is correctly set up for subsequent cloud computing lab exercises.